

DECODELABS

Industrial Training Kit — Batch 2026

Exploratory Data Analysis Report

Project 2 — EDA

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Role	Data Analytics Intern
Organization	DecodeLabs
Dataset	E-commerce Orders — 1,200 rows
Tools	Python , Pandas ,Matplotlib , Seaborn
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1. EXECUTIVE SUMMARY

This report presents the findings of an Exploratory Data Analysis (EDA) conducted on the DecodeLabs e-commerce orders dataset, containing 1,200 orders across a 2.5-year period (January 2023 – June 2025). The objective was to uncover hidden patterns, distributions, trends, and outliers to generate actionable business intelligence.

Problem Statement

which products drive the most revenue, what the customer's preferred payment method is, why only a fraction of orders are successfully delivered, which marketing channels attract the most customers, and when peak sales periods occur.

Methodology

- Data loaded and explored using Pandas (`df.head()`, `df.info()`, `df.describe()`)
- Missing values identified and handled — 309 nulls in `CouponCode` filled with 'No Coupon'
- Grouped aggregations performed using `groupby()` for product and time-based trends
- Visualizations produced using Matplotlib (bar, line, pie) and Seaborn (countplot, boxplot)
- Outlier detection using the IQR (Interquartile Range) method
- Correlation matrix computed to identify variable relationships

2. DATASET OVERVIEW

The dataset contains 1,200 rows and 14 columns covering e-commerce order transactions.

Column	Data Type	Description
OrderID	Text	Unique order identifier
Date	Date	Order date 2023–2025
CustomerID	Text	Unique customer
Product	Category	7 product types
Quantity	Integer	1–5 units per order
UnitPrice	Decimal	\$11.39–\$699.93
ShippingAddress	Text	Delivery address
PaymentMethod	Category	5 payment types
OrderStatus	Category	5 status types
TrackingNumber	Text	Unique shipment code
ItemsInCart	Integer	1–10 cart items
CouponCode	Category	3 codes + 309 nulls
ReferralSource	Category	5 marketing channels
TotalPrice	Decimal	Final order value

3. DESCRIPTIVE STATISTICS

Basic statistical summary of the three key numeric columns in the dataset.

Metric	Quantity	UnitPrice	TotalPrice
Count	1,200	1,200	1,200
Mean	2.95	\$356.41	\$1,053.97
Median	3.00	\$364.21	\$823.62
Std Dev	1.41	\$197.18	\$819.86
Min	1.00	\$11.39	\$11.39
Max	5.00	\$699.93	\$3,456.40

The Mean TotalPrice (\$1,053.97) is significantly higher than the Median (\$823.62), revealing a right-skewed distribution — a small number of high-value orders are pulling the average upward. Median is the more reliable measure of a 'typical' order for this dataset.

4. ANALYSIS & FINDINGS

Product Revenue Analysis

Revenue was summed per product using a groupby aggregation and visualized as a bar chart.

Product	Total Revenue	Observation
🏆 Chair	\$195,620.11	Top revenue generator
🥈 Printer	\$195,612.61	Almost equal to Chair
🥉 Laptop	\$192,126.56	Strong performer
Tablet	\$186,568.95	Mid-tier
Monitor	\$175,651.41	Mid-tier
Desk	\$167,459.93	Needs promotion
Phone	\$151,722.39	Lowest — requires strategy

Chair and Printer are the top revenue generators with almost identical totals (~\$195K each). Phone generates the least revenue at \$151,722 — nearly \$44K less than Chair. This gap suggests a targeted promotion or bundling strategy could significantly lift Phone revenue.

Payment Method Analysis

Order count was tallied per payment method to understand customer payment preferences.

Payment Method	Count	Share
Online	258	21.5% — Slight leader
Cash	246	20.5%
Credit Card	234	19.5%
Debit Card	232	19.3%
Gift Card	230	19.2%

Payment methods are remarkably evenly distributed (19–21% each). Online payment leads marginally. This balance indicates no single payment channel dominates, so the business should maintain support for all five methods without heavy investment in any one.

Order Status Analysis

Order counts grouped by status reveal the health of the fulfilment pipeline.

Order Status	Count	Share	Flag
Cancelled	250	20.8%	⚠ Critical
Returned	247	20.6%	⚠ High
Pending	237	19.8%	Monitor
Shipped	235	19.6%	In progress
Delivered	231	19.3%	✅ Success

CRITICAL FINDING: Only 19.3% of orders are Delivered. Cancelled (20.8%) and Returned (20.6%) orders together account for 41.4% of all orders — more than double the delivery success rate. This represents a severe business risk and requires immediate root-cause investigation across logistics, product quality, and customer experience.

Referral Source Analysis

Customer acquisition channels were analyzed by counting orders per referral source.

Referral Source	Count	Note
Instagram	259	Top channel — 21.6%
Email	250	Strong — 20.8%
Google	241	Organic — 20.1%
Facebook	228	19.0%
Referral	222	Word of mouth — 18.5%

Instagram drives the most orders (259), but the margin over other channels is very small. All five channels contribute nearly equally (18.5–21.6%), indicating a healthy multi-channel marketing strategy. Instagram should receive slightly more budget allocation as the top performer.

Monthly Sales Trend

Order dates were converted to monthly periods and total revenue was aggregated to identify seasonal patterns.

Month	Revenue	Note
June 2024	\$68,068.54	Peak month — highest revenue
May 2023	\$63,836.84	Second highest
January 2023	\$56,685.75	Strong start to 2023

June 2024 is the clear peak revenue month at \$68,068. The data spans 2023–2025, showing fluctuation across months. The business should plan promotional campaigns and increase inventory 6–8 weeks before the June peak to maximize the seasonal opportunity.

Outlier Detection (IQR Method)

The Interquartile Range method was applied to TotalPrice to identify orders that fall unusually far from the typical range.

Metric	Value	Meaning
Q1 (25th Pct)	\$410.52	Lower quartile
Q3 (75th Pct)	\$1,578.48	Upper quartile

IQR	\$1,167.96	Q3 - Q1
Upper Bound	\$3,330.42	Q3 + 1.5 × IQR
Outliers Found	8 orders	High-value — VIP signals

8 orders exceed the IQR upper bound of \$3,330.42. These are not data errors — they are genuine high-value orders (SIGNAL, not NOISE). These 8 customers represent potential VIP buyers and should be flagged for a loyalty or premium customer program to drive repeat purchases.

5. CORRELATION ANALYSIS

Pearson correlation coefficients were computed between numeric variables. Values above 0.6 (highlighted in green) indicate strong positive relationships.

	Quantity	UnitPrice	TotalPrice	ItemsInCart
Quantity	1.00	0.01	0.62	0.65
UnitPrice	0.01	1.00	0.72	0.00
TotalPrice	0.62	0.72	1.00	0.39
ItemsInCart	0.65	0.00	0.39	1.00

Variable Pair	Correlation	Business Meaning
UnitPrice → TotalPrice	0.72 (Strong)	Higher-priced items directly drive more revenue
Quantity → TotalPrice	0.62 (Strong)	Bulk orders significantly increase order value
Quantity → ItemsInCart	0.65 (Strong)	Customers who order more also browse more
UnitPrice → Quantity	0.01 (None)	Price does not affect how many units are ordered

Correlation ≠ Causation. A high correlation is a clue, not a verdict. Always investigate hidden variables before making decisions based purely on correlation values.

6. KEY FINDINGS & RECOMMENDATIONS

Summary of the 8 strongest signals discovered during the EDA, with actionable business recommendations.

	Finding	Recommended Action
1	Only 19.3% of orders are Delivered	Investigate fulfilment & logistics pipeline urgently
2	41.4% of orders Cancelled or Returned	Root-cause analysis on quality & delivery experience
3	Chair & Printer are top revenue products	Prioritize stock, ads, and shelf space for these
4	Phone generates the least revenue	Consider bundling, discounts, or promotional campaigns
5	June 2024 is peak revenue month	Plan seasonal campaigns 2 months before peak
6	Instagram is the top referral source	Increase Instagram marketing & influencer budget
7	8 high-value outlier orders detected	Identify VIP customers — build loyalty program
8	UnitPrice strongly correlates with Revenue	Focus on premium product listings and upsells

7. CONCLUSION

This Exploratory Data Analysis of 1,200 e-commerce orders revealed several critical business insights:

- Revenue is fairly balanced across all 7 products, with Chair and Printer as clear leaders
- The fulfilment pipeline is severely underperforming — only 19.3% of orders reach Delivered status
- All payment methods and referral channels are nearly equally distributed, showing business stability
- June 2024 is the peak revenue month, providing a clear seasonal planning target
- 8 high-value VIP customers were identified through IQR outlier analysis
- UnitPrice is the strongest predictor of total revenue ($r = 0.72$)

The most critical immediate action is investigating the 41.4% Cancelled + Returned order rate. Resolving this single issue could nearly double the effective delivery rate and significantly improve revenue retention.